

Automated Test Instruments Enable Seamless Working

For quicker measurement, control and data analysis in professional engineering labs

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The demand for test and measurement instruments to meet the lab infrastructure needs of professional engineering colleges is inevitably on the rise. To cater to various engineering fields offered, labs are equipped with a multitude of test and measurement instruments ranging from high-end oscilloscopes and spectrum analysers to common equipment such as digital multimeters and power supplies.

Traditionally, measurements are taken from a single instrument one at a time, recording the values. The measurement data is transferred to a PC to perform further data analysis. However, today, when measurements have become complex and integration of multiple instruments is required, this method is impractical as it is time-consuming and inefficient. Educators are therefore looking at integrating instruments into lab stations so that various tasks such as measurements, data analysis and data storage can be performed in a single PC.

Building an automated lab station

Building an automated instrument lab station requires consideration of three major components, namely, the software, hardware connectivity and instruments (see Fig. 1). The software on a PC is used to control the instrument that is connected to it using a connectivity hardware via an instrument control bus. Understanding each of these components helps in deciding

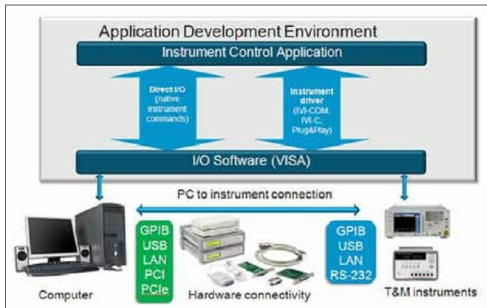


Fig. 1: Major components of an automated instrument lab station

which option is best suited for your application. Key considerations when building such a system are the ease of connectivity and scalability, performance and reliability.

Software. The first component to consider in designing an automated lab station is the software. It consists of three sub-components: The application development environment (ADE), instrument driver and input/output (I/O) software.

The ADE can either be a textual (C++, C# and VB.NET) or graphical programming language (National Instruments' LabVIEW and Agilent's VEE). In deciding which ADE is the most suitable for your application, factors such as the learning curve, ease of integration with other software and hardware, robust data analysis and management, reporting capabilities and support and maintenance costs

are important.

The instrument driver is essentially a set of software routines that allows quick communication with an instrument. Each routine corresponds to a programmed operation, such as configuring, writing to, reading from and triggering the instrument.

Two types of instrument drivers are prevalent today: Interchangeable virtual instrument (IVI) driver and plug-and-play drivers. The IVI standards specify an instrument driver architecture that provides compatibility with popular ADEs, common command syntax across instrument families and members and supports PC industry standards like COM (Component Object Model). The two architectures of IVI drivers are IVI-COM, which is based on Microsoft's COM standard, and IVI-C, which is based upon VXIplug&play specifica-